TEST REPORT

EN IEC 61496-3: 2019

Safety of machinery - Electro-sensitive protective equipment - Part 3: Particular requirements for active opto-electronic protective devices responsive to diffuse Reflection (AOPDDR)

Report Reference No......:	SHES220400619301-03
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Approved by (name + signature) .....:	Lucy Wang Lucy wf
Testing Laboratory.....:	SGS-CSTC Standards Technical Services (Shanghai) Co., Ltd.
Address .....:	588 West Jindu Road, Xinqiao, Songjiang, 201612 Shanghai, China.
Applicant's name .....:	Richbeam (Suzhou) Technology Co., Ltd.
Address .....:	No.788, Southeast Avenue, Changshu high tech Industrial Development zone, Jiangsu, China
Test specification	
Standard.....:	EN IEC 61496-3:2019
Test procedure.....:	SGS-CSTC
Master TRF.....:	N/A
Test item description .....:	2D TOF LIDAR
Trade Mark .....:	RICHBEAM
Manufacture.....:	Same as applicant
Model/Type reference .....:	LakiBeam 1, LakiBeam 1L
Ratings .....:	See nameplate

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Summary of testing:

All tests and assessments are performed on the prototype of the original sampled machine (LakiBeam 1L) and the technical file which are submitted by the client.
Considering the results of the performed test according to EN IEC 61496-3:2019, the items under test are IN COMPLIANCE with the requested specifications specified in the standard.

Tests performed (name of test and test clause):	**Test location:**
1. 5.2.1.2.2 Influence of incandescent light 2. 5.2.1.2.3 Influence of incandescent light reflected by the background 3. 5.2.1.3 Endurance test of the detection capability 4. 5.4.2.2 Ambient temperature variation and humidity test procedure 5. 5.4.2.2 Ambient temperature variation and humidity test procedure 6. 5.4.2.3 Condensing test procedure 7. 5.4.3.1 Supply voltage variations 5.4.3.2Supply voltage dips and interruptions 8. 5.4.3.3 Burst 9. 5.4.3.4 Surge 10. 5.4.3.5 Electromagnetic field 11. 5.4.3.6 Conducted disturbances induced by radio-frequency fields 12. 5.4.3.7 Electrostatic discharge 13. 5.4.4.1 Vibration 14. 5.4.4.1 Vibration 15. 5.4.4.2 Shock 16. 5.4.4.2 Shock 17. 5.4.4.3 Change of temperature 18. 5.4.4.4 Hammer tests 19. 5.4.5 Enclosures 20. 5.4.6.4 Light interference-Incandescent light 21. 5.4.6.5.2 Light interference-Fluorescent light 22. 5.4.6.5.3 Light interference-Fluorescent light (Fail to danger) 23. 5.4.6.6 Light interference-Flashing beacon 24. 5.4.6.7 Light interference-Laser beam pointer 25. 5.4.7.2 Pollution test with opaque test spot	No.1-31 test in Jining center for quality supervision and inspection of semiconductor& display products No. 32 test in Compliance Certification Services (Kunshan) Inc.

检验检测专用章 Inspection & Testing Services
SGS-CSTC Standards Technical Services (Shanghai) Co., Ltd. EEC Safety Lab

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26. 5.4.7.3 Test of homogeneous pollution of the emitter and receiver aperture 27. 5.4.8.1 Background interference 28. 5.4.9.1 Test with opaque test spots 29. 5.4.9.2 Manual interference test with AOPDDR covered 30. 5.4.9.3 Manual interference test in the zone with limited detection capability 31. 5.4.10 Optical occlusion in the detection zone (eclipsed by small object) 32. 5.4.3.7 Electrostatic discharge	

Copy of marking plate
(The artwork below may be only a draft.)



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Test item particulars..... :	
Classification of installation and use ..... :	Class I
Supply Connection ..... :	Fixed
Possible test case verdicts:	
- test case does not apply to the test object .....:	N/A
- test object does meet the requirement.....:	Pass
- test object does not meet the requirement .....:	Fail
Testing.....:	
Date of receipt of test item .....:	2022-04-29
Date (s) of performance of tests .....:	2022-05-10 to 2022-9-16

General remarks:

The test results presented in this report relate only to the object tested.

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"(see Enclosure #)" refers to additional information appended to the report.

"(see appended table)" refers to a table appended to the report.

Throughout this report, a point is used as the decimal separator.

Manufacturer's Declaration:	
The application for obtaining a Test Certificate includes more than one factory location and a declaration from the Manufacturer stating that the sample(s) submitted for evaluation is (are) representative of the products from each factory has been provided .....:	☐ **Yes** ☒ **Not applicable**

When differences exist; they shall be identified in the General product information section.

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Name and address of factory (ies) .....:	N/A

General product information and other remarks:

This product is 2D TOF LIDAR.
Ranging type radar, no OSSDs.
In the test, monitoring the output distance instead of the status of the OSSD.

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EN IEC 61496-3			
Clause	Requirement + Test	Result - Remark	Verdict
4	**Functional, design and environmental requirements**		—
4.1	Functional requirements		—
4.1.3	Types of ESPE		-
	Replacement: In this document, only type 2 and type 3 ESPE are considered. The types differ in their performance in the presence of faults and under influences from environmental conditions. It is the responsibility of the machine supplier and/or the user to prescribe which type is suitable for a particular application.	Type 1 ESPE	N/A
	The type 2 ESPE shall fulfil the fault detection requirements of 4. 2. 2. 3 of this document. In normal operation, the output circuit of each of at least two output signal switching devices (OSSDs) or of one output signal switching device (OSSD) and one secondary switching device (SSD) of the type 2 ESPE shall go to the OFF-state when the sensing device is actuated, or when the power is removed from the device.	Type 1 ESPE	N/A
	The type 3 ESPE shall fulfil the fault detection requirements of 4. 2. 2.4 of this document. In normal operation, the output circuit of each of at least two output signal switching devices (OSSDs) of the type 3 ESPE shall go to the OFF-state when the sensing device is actuated, or when the power is removed from the device.	Type 1 ESPE	N/A
	When a single safety- related data interface is used to perform the functions of the OSSD(s) , then the data interface and associated safety-related communication interface shall meet the requirements of 4. 2.4.4. In this case, a single safety- related data interface can substitute for two OSSDs in a type 3 ESPE.	Type 1 ESPE	N/A
	Addition:		-
4.1.6	Zone(s) with limited detection capability		-

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Clause	Requirement + Test	Result - Remark	Verdict
	In order to ensure no hazard can arise in a particular application due to the presence of one or more zone(s) with limited detection capability between the optical window and the detection zone, its dimensions and appropriate information for use shall be provided by the supplier.		P
	If the zone with limited detection capability extends more than 50 mm from the optical window in direction to the detection zone(s) , then additional and effective technical measures shall be applied to prevent undetected presence of objects or persons or parts of persons in the zone with limited detection capability.	No this situation	N/A
4.2	Design requirements		-
4.2.2	Fault detection requirements		-
4.2.2.2	Particular requirements for a type 1 ESPE		-
	4.2.2.2 of IEC 61496-1:2012 does not apply.	Type 1 ESPE	P
4.2.2.3	Particular requirements for a type 2 ESPE		-
	Replacement:		-
	A type 2 ESPE shall have a means of periodic test to reveal a failure to danger (for example loss of detection capability, response time exceeding that specified).	Type 1 ESPE	N/A
	The test shall be performed also at power-on of the ESPE before going to the ON-state and at each reset.		N/A
	Depending on the application, the periodic test may need to be performed more often to achieve a desired safety performance. Generic functional safety standards give requirements how often periodic test have to be applied to fulfil the requirements for a certain safety performance.		N/A
	NOTE 1 The periodic test can be initiated by external or internal means.		-
	When it is not possible to reveal a failure to danger by periodic tests other equivalent measures shall be applied.		N/A

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EN IEC 61496-3			
Clause	Requirement + Test	Result - Remark	Verdict
	A single fault resulting in the loss of the stated AOPDDR detection capability or the increase in response time beyond the specified time or preventing one or more of the OSSDs going to the OFF-state shall result in a lock-out condition as a result of the next periodic test.		N/A
	A single fault resulting in the deterioration of the stated AOPDDR detection capability shall result in a lock-out condition at least as a result of the next periodic test. If periodic test cycle is less than 5 s then deterioration of the stated AOPDDR detection capability shall be detected within 5 s.		N/A
	NOTE 2 Examples of deterioration of the AOPDDR detection capability include – the increase of the minimum detectable object size, – the increase in the minimum detectable reflectivity, and; – the decrease of position accuracy.		-
	The occurrence of single faults shall be considered by analysis and/or test with each of the following conditions and throughout the entire detection zone:		N/A
	– environmental conditions specified in 4.3;		N/A
	– at the limits of alignment and/or adjustment.		N/A
	Where the periodic test is intended to be initiated by an external (for example machine) safety-related control system, the ESPE shall be provided with suitable input facilities (for example terminals).		N/A
	The duration of the periodic test shall be such that the intended safety function is not impaired, especially if the ESPE is intended for use as a trip device.		N/A
	If the periodic test is automatically initiated, the correct functioning of the periodic test shall be monitored. In the event of a fault, the OSSD(s) shall be signalled to go to the OFF-state. If one or more OSSDs do(es) not go to the OFF-state, a lock-out condition shall be initiated.		N/A
	An ESPE with only one OSSD shall have a minimum of one SSD (see Clause A.4 of IEC 61496-1:2012).		N/A

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EN IEC 61496-3			
Clause	Requirement + Test	Result - Remark	Verdict
4.2.2.4	Particular requirements for a type 3 ESPE		-
	Replacement:		
	A single fault in the sensing device resulting in a complete loss of the stated AOPDDR detection capability shall cause the ESPE to go to a lock-out condition within the specified response time.		N/A
	NOTE 1 For AOPDDR using rotating mirrors for scanning the detection zone, this requirement can be fulfilled by scanning on a defined reference object located outside the detection zone and the tolerance zone.		N/A
	A single fault resulting in a deterioration of the stated AOPDDR detection capability shall cause the ESPE to go to a lock-out condition within a time period of 5 s following the occurrence of that fault.		N/A
	NOTE 2 Examples of deterioration of the AOPDDR detection capability include – the increase of the minimum detectable object size, – the increase in the minimum detectable reflectivity, and – the decrease of position accuracy.		-
	A single fault resulting in an increase in response time beyond the specified value or preventing at least one OSSD going to the OFF-state shall cause the ESPE to go to a lock-out condition within the response time, or immediately upon any of the following demand events where fault detection requires a change in state:		N/A
	– on actuation of the sensing function;		N/A
	– on switch off/on;		N/A
	– on reset of the start interlock or the restart interlock, if available (see Clauses A.5 and A.6 of IEC 61496-1:2012);		N/A
	– on the application of an external test signal, if available.		N/A
	An external test signal can be required if, for example, in a particular application, the frequency of actuation of the sensing function is foreseeably low and the OSSDs are monitored only at the change of state.		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	In cases where a single fault which in itself does not cause a failure to danger is not detected, the occurrence of one additional fault shall not cause a failure to danger. For verification of this requirement, see 5.3.4.		N/A
	The occurrence of single faults shall be considered by analysis and/or test with each of the following conditions and throughout the entire detection zone:		N/A
	– environmental conditions specified in 4.3;		N/A
	– at the limits of alignment and/or adjustment.		N/A
4.2.2.5	Particular requirements for a type 4 ESPE		-
	4.2.2.5 of IEC 61496-1:2012 does not apply.		N/A
	Addition:		-
4.2.12	Integrity of the AOPDDR detection capability		-
4.2.12.1	General		-
	The design of the AOPDDR shall ensure that the detection capability is not decreased below the limits specified by the supplier and in this document by any of, but not limited to, the following:		N/A
	a) reflectivity of objects in the range defined for the test pieces to be detected;		N/A
	b) the position, size and number of objects within the detection zone;		N/A
	c) the size of detection zones;		N/A
	d) auto-adjustment, for example the following: 1) gain control; 2) sample rate; 3) shutter time; 4) optical characteristics;		N/A

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	e) properties/limitations of the emitting/receiving element, optics and signal processing, for example the following: 1) signal noise; 2) dynamic range; 3) sensitivity and uniformity (e.g. cold and hot pixels); 4) micro lenses; 5) change of characteristics;		N/A
	f) calibration of the sensing device;		N/A
	g) accuracy of object position in image(s);		N/A
	h) at the limits of alignment and/or adjustment;		N/A
	i) environmental conditions specified in 4.3;		N/A
	j) component tolerances;		N/A
	k) changing of characteristics of internal and external references to ensure the detection capability.		N/A
	NOTE 1 Under certain circumstances, limitations of the sensor in relation to its use need to be considered. For example, – objects that generate mirror-like (specular) reflections cannot be detected if the portion of diffuse reflectivity is less than that specified for the "black" test piece; – the determination of the minimum reflectivity for the detection of obstacles is based on the clothing of a person; it is possible that objects having a reflectivity lower than that considered in this document are not detected.		-
	NOTE 2 The technique of scanning on a reference object can satisfy the requirement in respect of ageing of components. Other techniques giving the same level of assurance can be used.		-
4.2.12.2	Detection zone(s) and tolerance zone(s)		-
	The supplier shall specify the tolerance zone(s).		P
	The supplier shall take into account worst-case conditions including, for example, signal-tonoise ratio S/N and standard deviation σ considering all influences listed in this document and any additional influences specified by the supplier (environmental influence, component faults, multi-path reflections etc.).		P

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Clause	Requirement + Test	Result - Remark	Verdict
	The supplier shall specify the relevant parameters of the detection zone(s), including operating distance and scanning angle or field of view. The geometry and/or frequency shall be sufficient to ensure that a test piece with a diameter of the specified minimum detectable object size is detected at the maximum operating distance. The supplier shall define values in the range of 30 mm to 200 mm as the minimum detectable object size of the AOPDDR. The minimum detectable object size may be distance dependent.		P
	The restriction of the minimum detectable object size to the range of 30 mm to 200 mm is based on current applications. Additional requirements can be necessary for AOPDDRs having detection capabilities outside this range.		P
	NOTE 1 For an AOPDDR-2D the detection capability can be determined by the optical geometry so that one complete beam will impinge on the specified test pieces in the maximum distance of detection zone and tolerance zone for a special design. In this case, the distance between the centre of two adjacent emitter beams (except the first and the last one) will not exceed half the diameter of the test pieces. For other designs, it can be more difficult to carry out the verification according to 5.2.1.2 and 5.2.11, especially when movement of objects is taken into account.		-
	All points on a path projected from any point on the border of the detection zone to the receiving element(s) of the AOPDDR shall be within the detection zone or the zone with limited detection capability (see 4.1.6).		P
	The tolerance zone depends on systematic influences, measurement faults, resolution of the measurement values, etc. and is necessary to ensure the required detection probability within the detection zone. Figures 1 and 2 show examples of tolerance zones.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	The test pieces (see 4.2.13) shall be detected with a minimum probability of detection of 1 – 2,9 × 10–7 throughout the detection zone(s). To achieve this minimum probability of detection, the tolerance zone is added to the detection zone (see Figure BB.2). Even if a measured distance value of a test piece falls into the tolerance zone, this test piece will be determined as detected and the OSSDs shall go to the OFF-state or remain in the OFF-state.		P
	NOTE 2 The tolerance zone is not included in the detection zone.		-
	NOTE 3 Probability of detection as used in this document is not related to the probability of faults.		-
	Special attention may be required when the detection zone of AOPDDR is made up by more than one emitting and/or receiving unit to ensure that the AOPDDR detection capability is not affected between the fields of view of these units.	one emitting and receiving unit	N/A
	When a test piece is placed on the boundary between the detection zone and the tolerance zone (i.e. on the border of the detection zone) the measured distance values of this test piece shall be the median point of the distribution of measurement values determined using a test piece with a reflectivity of any value from that of the "black" test piece to that of the "white" test piece. The supplier shall document the reflectivity of the test piece and the calculations used.	Report of test piece provided	P
	The value for the position accuracy and the tolerance zone is not necessarily a constant. It can, for example, be a function of the measurement distance. It can be necessary to take into account that the size of parts of the tolerance zone can be related for example to the diameter of the test piece and the beam position. The dimensions of the tolerance zone in the three axes can be different.		P
	NOTE 4 If the AOPDDR has a facility to automatically set its detection zone(s), the ranging error of the set values is taken into account when determining the tolerance zone (see Clause A.11).		-
	NOTE 5 Annex BB gives additional information about the relationship between position accuracy and probability of detection.		-
4.2.12.3	Influences on detection		-

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Clause	Requirement + Test	Result - Remark	Verdict
4.2.12.3.1	General		-
	Objects at the limits of the detection capability that are either stationary or moving within the detection zone at any speed up to 1,6 m/s shall be detected by the ESPE within the specified response time. The response time shall be determined by the supplier taking into account worst-case conditions and the movement of objects. Where the supplier states that an AOPDDR can be used to detect objects moving at speeds greater than 1,6 m/s, the requirements shall be met at any speed up to and including the stated maximum speed(s).		P
	NOTE An example for the calculation of the response time is given in Clause AA.6.		-
4.2.12.3.2	AOPDDR used as a trip device with orthogonal approach (200 mm minimum detectable object size)		N/A
	When using an AOPDDR as a trip device with 200 mm minimum detectable object size, the height of the detection zone above the reference plane, for example the floor, shall include the range between 300 mm and 1 400 mm. Figure 3 shows an example.		N/A
	NOTE 1 Aspects of reaching over are dealt with in ISO 13855.		-
	The sensing device shall be actuated and the OSSDs shall go to the OFF-state when a cylindrical test piece with a diameter of 200 mm and a length of 200 mm in accordance with 4.2.13 is moving through the detection zone such that the direction of movement and the axis of the test piece are orthogonal to the detection zone at a speed of 1,6 m/s.		N/A
	Where the supplier states that an AOPDDR can be used to detect objects moving at speeds greater than 1,6 m/s, this requirement shall be met at the stated maximum speed. Where the supplier states a minimum detectable object size smaller than 200 mm for the use of an AOPDDR as a trip device with orthogonal approach, this value shall be used as the diameter of the cylindrical test piece.		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	When the OSSD(s) go to the OFF-state, they shall remain in the OFF-state while the test piece is present in the detection zone.		N/A
	NOTE 2 The purpose of this requirement is to ensure that the OSSDs go to the OFF-state when a person passes through the detection zone.		-
4.2.12.3.3	AOPDDR used as a trip device with orthogonal approach (150 mm minimum detectable object size)		-
	When using an AOPDDR as a trip device with 150 mm minimum detectable object size the height of the detection zone above the reference plane, for example the floor, shall include the range between 300 mm and 1 100 mm. Figure 4 shows an example.		N/A
	NOTE 1 Aspects of reaching over are dealt with in ISO 13855.		-
	The sensing device shall be actuated and the OSSDs shall go to the OFF-state when a cylindrical test piece with a diameter of 150 mm and a length of 150 mm in accordance with 4.2.13 is moving through the detection zone such that the direction of movement and the axis of the test piece are orthogonal to the detection zone at a speed of 1,6 m/s.		N/A
	Where the supplier states that an AOPDDR can be used to detect objects moving at speeds greater than 1,6 m/s, this requirement shall be met at the stated maximum speed. Where the supplier states a minimum detectable object size smaller than 150 mm for the use of an AOPDDR as a trip device with orthogonal approach, this value shall be used as the diameter of the cylindrical test piece.		N/A
	When the OSSD(s) go to the OFF-state, they shall remain in the OFF-state while the test piece is present in the detection zone.		N/A
	NOTE 2 The purpose of this requirement is to ensure that the OSSDs go to the OFF-state when a lower limb of a person passes through the detection zone.		-
4.2.12.4	Minimum detection zone		-
	The supplier shall specify the minimum detection zone(s). The supplier shall take into account the conditions as described in 4.2.12.3.1.		P

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Clause	Requirement + Test	Result - Remark	Verdict
4.2.13	Test pieces for type testing		-
4.2.13.1	General		-
	The test pieces are part of the AOPDDR and shall therefore be provided by the supplier for use in the type tests of Clause 5. They shall be marked with a type reference and identification of the AOPDDR with which they are intended to be used.		P
	For AOPDDR-2D, the test pieces shall be a cylinder with a diameter equal to the minimum detectable object size and a minimum effective length of 0,2 m except the dimensions which are defined in 4.2.12.3.2 and 4.2.12.3.3. Other diameters within the range of 30 mm to 200 mm may be required for testing depending on the minimum detectable object size of the AOPDDR-2D.		P
	NOTE 1 The minimum effective length of the test pieces has been selected for ease of use.		-
	For AOPDDR-3D, the test pieces shall be cylinders as defined for AOPDDR-2D or shall be a conical test piece – see a) – and a cylindrical test piece – see b) and c).	2D AOPDDR	N/A
	a) Conical test piece: The test piece shall be a truncated cone if the AOPDDR-3D is intended to be used for hand detection. The test piece does start with a diameter of 20 mm increasing up to 40 mm over a length of 160 mm.		N/A
	The test piece shall be a truncated cone in combination with a cylinder if the AOPDDR-3D is intended to be used for arm detection. The test piece starts with a diameter of 40 mm increasing up to 55 mm as a cone over a length of 180 mm and continues as a cylinder with a diameter of 55 mm to an overall length of 440 mm.		N/A
	The test piece shall be a truncated cone if the AOPDDR-3D is intended to be used for leg detection. The test piece starts with a diameter of 50 mm increasing up to 117 mm over a length of 1 000 mm.		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	If the AOPDDR-3D is intended to be used for detection of different parts of a body, the selection of the most appropriate test pieces shall be dependent on the analysis of the design and intended application. In some cases, all test pieces can be required.		N/A
	b) Cylindrical test piece: If the AOPDDR-3D is intended to be used for lower limb detection, then the test piece shall be a cylinder with a diameter of 150 mm and a length of 150 mm.		N/A
	c) Cylindrical test piece: If the AOPDDR-3D is intended to be used for whole body detection, then the test piece shall be a cylinder with a diameter of 200 mm and a length of 200 mm.		N/A
	NOTE 2 A cylindrical test piece with a diameter of 200 mm is intended to represent the thickness of a body.		-
4.2.13.2	Black test piece		-
	The surface of the test piece shall have a coefficient of diffuse reflection in the range of 1,6 % to 2,0 % including measurement accuracy, at the emitted wavelength under normal conditions. This value shall be verified by measurement. Where this reflectivity is used for calculation, the nominal value of 1,8 % shall be used.		P
	Figure 5 shows the results of an investigation to determine the reflectivity of the black test piece (performed by Institut für Arbeitsschutz (IFA), 53757 Sankt Augustin, Germany).		P
4.2.13.3	White test piece		-
	The surface of the test piece shall have a coefficient of diffuse reflection in the range of 80 % to 90 % at the emitted wavelength.		P
4.2.13.4	Retro-reflective test piece		-
	The surface of the test piece shall be of retro-reflecting material. The material shall comply with the requirements for separate performance retro-reflective material of ISO 20471.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	NOTE Table 4 of ISO 20471:2013 defines the minimum coefficient of retro-reflection for separate performance retro-reflective material as 330 cd · lx–1 · m–2 with an entrance angle of 5° and an observation angle of 0,2° (12').		-
4.2.14	Wavelength		-
	AOPDDRs shall operate at a peak wavelength in the range of 820 nm to 950 nm.	940nm	P
	NOTE This range of wavelengths is based on the present availability of components together with research which shows it to be suitable for materials used as clothing.		-
4.2.15	Radiation intensity		-
	If the emitting element(s) uses LED technology, the radiation intensity generated and emitted by the AOPDDR shall meet the requirements of exempt group in accordance to IEC 62471.	No LED emitting element	N/A
	NOTE Exempt group is equal to risk group zero (IEC 62471:2006).		-
	If the emitting element(s) uses laser technology, the radiation intensity generated and emitted by the AOPDDR shall not exceed the maximum power or energy levels for a class 1M laser in accordance with IEC 60825-1 (even in the presence of a component failure). The marking as a class 1 or class 1M laser shall be carried out as required in 5.2 of IEC 60825-1:2014.	Class 1 laser product	P
4.2.16	Mechanical construction		-
	When the detection capability can be decreased below the limit stated by the supplier, as a result of a change of position of components, the fixing of those components shall not rely solely on friction.		P
	NOTE The use of oblong mounting holes without additional means could lead for example to a change of the position of the detection zone under mechanical influence such as shock.		-
4.3	Environmental requirements		-
	4.3 of IEC 61496-1:2012 applies, except as follows.	Tested	P
	Addition:		-
	NOTE It is possible that these requirements do not fulfil the needs of certain applications (for example: use on vehicles, including automatic guided vehicles (AGVs), forklifts, mobile machinery, etc.).		-

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Clause	Requirement + Test	Result - Remark	Verdict
4.3.1	Ambient air temperature range and humidity		-
	4.3.1 of IEC 61496-1:2012 applies, except as follows.	Tested	P
	Addition:		-
	The ESPE shall not fail to danger when subjected to a rapid change of temperature and humidity leading to condensation on the optical window.	Tested	P
	This requirement is verified by the condensing test of 5.4.2.	Tested	P
4.3.3	Mechanical environment		-
4.3.3.1	Vibration		-
	Replacement:		-
	The ESPE shall pass the vibration tests of 5.4.4.1.	Tested	P
	Replacement, including the title of 4.3.3.2 of IEC 61496-1:2012:		P
4.3.3.2	Shock		-
	The ESPE shall pass the shock tests of 5.4.4.2.	Tested	P
	Addition:		-
4.3.3.3	Change of temperature		
	The ESPE shall be free of damage, including displacement and/or cracks of the optical window, after the tests of 5.4.4.3 and it shall be capable of continuing in normal operation.	Tested	P
4.3.3.4	Impact resistance		-
4.3.3.4.1	Normal operation		-
	The ESPE shall be free of damage, including displacement and/or cracks of the optical window, after the tests of 5.4.4.4.2 and it shall be capable of continuing in normal operation.	Tested	P
4.3.3.4.2	Fail to danger		-
	The ESPE shall not fail to danger after the tests of 5.4.4.4.3.	Tested	P

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Clause	Requirement + Test	Result - Remark	Verdict
4.3.4	Enclosures		-
	4.3.4 of IEC 61496-1:2012 applies, except as follows.	Tested	P
	Addition:		-
	Means shall be provided for the secure fixing of the enclosure(s).		P
	Enclosures of the AOPDDR containing optical components shall provide a degree of protection of at least IP65 (see IEC 60529) when mounted as specified by the supplier.		P
	Addition:		-
4.3.5	Light interference on AOPDDR receiving elements and other optical components		-
	The ESPE shall continue in normal operation when subjected to the following:	Tested	P
	– incandescent light;		P
	– fluorescent light operated with high-frequency electronic power supply;		P
	– radiation from an AOPDDR of identical design if no mounting restrictions related to possible interference are given by the supplier of the AOPDDR.		P
	The ESPE shall not fail to danger when subjected to the following:		-
	– high-intensity incandescent light;		P
	– fluorescent light operated with its rated power supply and with a high-frequency electronic power supply;		P
	– radiation from an AOPDDR of identical design;		P
	– flashing beacon;		P
	– collimated laser beam;		P
	These requirements are verified by the tests of 5.2.1.2 and 5.4.6.	Tested	P
4.3.6	Pollution interference		-

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Clause	Requirement + Test	Result - Remark	Verdict
	The supplier shall specify the maximum level of homogeneous pollution in percentage of transmission which will not result in a decrease of the stated detection capability.	Tested	P
	The AOPDDR shall continue in normal operation when the received signal energy of the detection system itself is attenuated by up to 30 % by homogeneous pollution.		P
	Pollution between the emitting and/or receiving element(s) and the beginning of the detection zone(s) (including optical components) of the AOPDDR resulting in a loss of the stated detection capability shall cause the OSSDs to go to the OFF-state.		P
	These requirements are verified by the tests of 5.4.7.		P
	NOTE It is possible that the tests listed in 5.4.7 do not cover all possible forms of pollution, for example, oil, grease and process materials.		-
	Any pollution monitoring means for detecting a loss of the stated detection capability shall comply with all the relevant requirements of this document.		P
4.3.7	Interference by surrounding objects		-
4.3.7.1	Background interference		-
	The stated tolerance zone shall not be increased by background interference. This requirement is verified by the tests of 5.4.8.1.	Tested	P
	NOTE 1 The supplier can specify the AOPDDR for a maximum reflectivity that is monitored by the AOPDDR itself and which leads to the OFF-state of the OSSDs if the specified maximum reflectivity is exceeded. Background interference by materials with higher reflectivity can thereby be excluded.		-
	NOTE 2 Backgrounds that can interfere with the measurement results include corner cube reflectors, tiles, sheet metal, white paper, etc.		-
	NOTE 3 Retro-reflectors are considered as a background within the tests of detection capability and position accuracy (see 5.4.8.1). If retro-reflectors in the background lead to measurement faults, it can be possible in specific applications to use other measures instead of an addition to the tolerance zone.		-
4.3.7.2	Multi-path reflections		-

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Clause	Requirement + Test	Result - Remark	Verdict
	The stated tolerance zone shall not be increased by multi-path reflections. This requirement is verified by the tests of 5.4.8.2.		N/A
4.3.8	Manual interference		-
	It shall not be possible to reduce the stated detection capability by covering the optical window of the housing of the AOPDDR or other parts (if applicable) or by placing objects within a zone with limited detection capability (see 4.1.6). In such cases, the OSSDs shall go to the OFF-state within a time period of 5 s and the OSSDs shall remain in the OFF-state until the manual interference is removed.		P
	AOPDDR for use as trip device with orthogonal approach shall be designed such that the OSSDs shall go to the OFF-state within the stated response time when manual interference leading to a reduction of the stated detection capability is performed and the OSSDs shall remain in the OFF-state until the manual interference is removed.		P
	These requirements are verified by the tests of 5.4.9.		P
4.3.9	Optical occlusion in the detection zone (eclipsed by small object)		-
	The AOPDDR detection capability shall be maintained when small objects are present in the detection zone. This shall be verified by analysis and by a test according to 5.4.10. The analysis shall include examination of any software filtering algorithms provided.	Tested	P
	NOTE Software filtering algorithms can be provided to disregard small objects, for example, to increase reliability of operation.		-
4.3.10	Drift or ageing of components		-
	For a type 2 ESPE, drift or ageing of components that would reduce the detection capability below the value stated shall not cause a failure to danger of the ESPE and shall be detected at least at the next periodic test and shall lead to an OFF-state. If periodic test cycle is less than 5 s then drift or ageing of components shall be detected within 5 s.	Type 1 ESPE	N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	For a type 3 ESPE, drift or ageing of components that would reduce the detection capability below the value stated shall not cause a failure to danger of the ESPE and shall be detected within a time period of 5 s and shall lead to an OFF-state.	Type 1 ESPE	N/A
	If a reference object is used for monitoring ageing and drift of components, variations in its properties (for example reflectivity) shall not cause a failure to danger of the ESPE. If a reference object is used to monitor ageing and drift of components, it shall be considered to be part of the AOPDDR and shall be provided by the supplier of the AOPDDR.		N/A
5	Testing		-
	This clause of part 1 is applicable except as follows:		P
5.1	General		-
5.1.1.2	Operating condition		-
	5.1.1.2 of IEC 61496-1:2012 applies, except as follows.		P
	Addition:		-
	As a result of the analysis of the design and the integrity of the detection capability of the AOPDDR, a test plan shall be established considering the test conditions and parameters outlined in this document. In the following tests, it shall be verified that when the OSSDs go to the OFF-state, they remain in the OFF-state while the test piece is present in the detection zone.		P
	The minimum test conditions shall be as specified in this document or by the supplier, whichever is more stringent.		P
	Tests can be omitted if an analysis can demonstrate that	No this situation	N/A
	- limited intended use in application justifies omission of test (e.g. only top mounting), and		N/A
	- a less stringent test is replaced by a more stringent test (e.g. test of equal detection capability at minimum working range versus maximum working range).		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	The analysis shall be documented in the test plan.		N/A
	Unless otherwise stated in this document, and if the facility is provided to set the detection zone, the zone used for the tests shall be set up as follows:		N/A
	- radius, respectively width and length (or equivalent values), of the detection zone of 1,0 m, called basic test distance (BTD);		N/A
	- add the value of the specified tolerance zone.		N/A
	NOTE For example, a detection zone of 1,0 m and a tolerance zone of 0,2 m results in a zone used for the tests of 1,2 m.		-
	For an AOPDDR with a stated maximum detection distance of less than 1,0 m, this maximum distance shall be used as BTD. For an AOPDDR with a stated minimum detection distance of greater than 1,0 m, this minimum distance shall be used as BTD.	maximum detection distance is 15m	N/A
	For an AOPDDR without the facility to set the detection zone, the fixed detection zone shall be used for all tests.		P
	All tests shall be performed with the test piece axis orthogonal to the detection zone for an AOPDDR-2D. For an AOPDDR-3D, all tests shall be performed with the test piece axis orthogonal to the centre axis or corner axis. Tests at other orientations of the test piece may be required depending on analysis of the design and worst-case considerations.		P
5.1.2.2	Measurement accuracy		-
	Addition to first paragraph:		-
	- for precision of distance/position measurements: 10 % of the stated position accuracy;	Tested	P
	- for light intensity measurement: ±10 %.	Tested	P
5.2	Functional tests		-
5.2.1	Sensing function		-
	Replacement:		-
5.2.1.1	General		-

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Clause	Requirement + Test	Result - Remark	Verdict
	The sensing function and the integrity of the detection capability shall be tested as specified, taking into account the following.		P
	a) The systematic analysis required in 5.2.1.2.1.	Tested	P
	b) The tests shall verify that the specified test pieces are detected when the test piece is either static or moving into or within the configured detection zone at any speed from 0 m/s to 1,6 m/s. Where the supplier states that objects can be detected moving at higher speeds, the requirements shall be met at all speeds up to the stated maximum speeds.		P
	NOTE 1 The AOPDDR can detect a test piece moving at any speed from 0 m/s to 1,6 m/s only when it remains inside the configured detection zone for at least the minimum response time and other influences on the minimum detection zone (see 4.2.12.4). The detection zone can be adjusted to the minimum detection zone for testing.		-
	c) The tests shall verify that the specified test pieces are detected when the test piece is placed inside the detection zone(s) by half of the minimum detectable object size for AOPDDR-2D and placed inside the detection zone(s) as far as the minimum detectable object size for AOPDDR-3D (see for example Figure 6).		P
	NOTE 2 The different test piece positions for AOPDDR-2D and AOPDDR-3D are due to the following: - backwards compatibility to IEC 61496-3:2008 for AOPDDR-2D; because of the values of half of the detection capability, the effect in practice is negligible in most cases; - AOPDDR-3D can usually detect a test piece with the required probability of detection only when the test piece is completely inside the detection zone.		-
	d) The tests shall verify the dimension(s) of the tolerance zone (i.e. the position accuracy) stated by the supplier.		P
	e) The number, selection and conditions of the individual tests shall be such as to verify the requirements of 4.2.12.1.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	It shall be verified that the sensing device is continuously actuated and, where appropriate, that the OSSD(s) go to the OFF-state as described in the tests below, taking into account the operating principle of the AOPDDR and, in particular, the techniques used to provide tolerance to environmental interference.		P
	Table 1 shows an overview of the minimum tests required for the verification of detection capability requirements.		P
	For AOPDDR-3D the tests according to 5.2.1.2 and 5.2.10 listed in Table 1 shall be carried in a minimum of two different distance sets:		P
	- distance values according to Table 1 through the centre;		P
	- distance values according to Table 1 along the corner.		P
	All other tests listed in Table 1 shall be carried out through the centre for AOPDDR-3D.		P
	If a distance between the origin of distance measurement at the AOPDDR and the test piece given in Table 1 is not possible, for example when BTD is less than 0,5 m, then the related tests can be omitted.		N/A
5.2.1.2	Integrity of the detection capability		-
5.2.1.2.1	General		-
	It shall be verified that the stated AOPDDR detection capability is maintained or the ESPE does not fail to danger, by systematic analysis of the design of the AOPDDR, using testing where appropriate and/or required, taking into account the conditions specified in 4.2.12.1 and the faults specified in 5.3.3, respectively 5.3.4. The results of this systematic analysis shall identify which tests in Clause 5 require, in addition, a measurement of the response time.		P
	NOTE 1 For details of fault combinations see 5.3.		-

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Clause	Requirement + Test	Result - Remark	Verdict
	The conditions and the number of measurements required to determine the integrity of the detection capability shall take into account the objectives of 5.2.1.1. As a minimum, the series of measurements listed in Table 1 and Table 2 shall be carried out at each position necessary to verify the integrity of detection capability within the detection zone. For AOPDDRs with more than one emitting and/or receiving element, it may be necessary to carry out measurements for each element. When measurement values are required for verification, each test result shall be based on a minimum of 1 000 single measurements at each position of the test piece.		P
	The use of special tools supplied by the supplier can be necessary to perform certain tests involving the recording and analysis of measurement values.		P
	The test arrangement used for the tests of 5.2.1.2.2 and 5.2.1.2.3 shall be compatible with the characteristic of the AOPDDR under test. The light interference tests shall be carried out at least with the "black“ test piece (see 4.2.13.2) at the BTD between the origin of distance measurement and test piece and the maximum operating distance - see also 5.2.1.1 d). The test sequence for the light interference tests shall be as follows:		P
	- the test piece shall be placed at the required distance before the test starts which, for tests according to Figure 8, is the border of the detection zone;		P
	- the start or restart interlock shall not be operational whilst the tests according to Figure 8 are performed;		P
	- AOPDDR shall be in normal operation and OSSDs in the OFF-state whilst the tests according to Figure 8 are performed;		P
	- the interfering light source shall then be switched on;		P
	- the test shall be continued for a time period of 3 min.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	Due to the inherent design of the AOPDDR, for example, the opto-mechanical construction, it can be necessary to carry out an extra series of measurements at additional distances.		P
	NOTE 2 Diagnosis and configuration tools (for example, software) belonging to the AOPDDR can be used for these measurements.		-
5.2.1.2.2	Influence of incandescent light		-
	The influence of incandescent light on the integrity of the detection capability shall be tested using the configuration shown in Figures 7 or 8 at the BTD between origin of distance measurement and test piece and the maximum operating distance - see also 5.2.1.1 d). When testing according to Figure 7, measurement values are required to verify the integrity of the detection capability. When testing according to Figure 8, the ESPE shall stay in the OFFstate during the test sequence.	Tested	P
	The measurement of the light intensity shall be carried out at the optical window of the AOPDDR when testing with an operating distance of 1,0 m. When testing at the maximum operating distance, the measurement of the light intensity shall be carried out in the detection zone at a distance of 1,0 m from the test piece towards the AOPDDR. The interfering light shall be directed along the optical axis of one or more receiving element(s). The test for the influence of incandescent light on the integrity of the detection capability shall be performed as follows:		P
	- The light shall be directed along the centre axis of an AOPDDR.		P
	- The light intensity shall be as close as possible to a maximum value of 3 000 lx consistent with the AOPDDR remaining in normal operation.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	- If the highest level of direct illumination with which the AOPDDR remains in normal operation is below 1 500 lx, an additional test shall be carried out with light being reflected to the AOPDDR by an object measuring 0,5 m × 0,5 m and having a diffuse reflective surface. The object shall be located outside the detection zone and the tolerance zone. The coefficient of diffuse reflection of the object used for this test shall be greater than 80 % in the range of wavelengths used by the AOPDDR and in the range used for the measurement of intensity. The light intensity for this additional test shall be as close as possible to a maximum value of 3 000 lx consistent with the AOPDDR remaining in normal operation.		P
	NOTE The relative position of the interfering light source, the test piece and the AOPDDR can affect the detection capability. For example, loss of detection capability due to the existence of a recovery time can be revealed when scanning the test piece immediately after the interfering light source.		-
	For AOPDDR-2D the set-up has to be adapted accordingly, see 5.2.1.1 c).		P
	NOTE Figure 8 shows a possible configuration for a test according to 5.2.1.2.2.		-
5.2.1.2.3	Influence of incandescent light reflected by the background		-
	The influence on the integrity of the detection capability by incandescent light reflected by the background shall be tested using the configuration shown in Figure 9 at the BTD between origin of distance measurement and test piece and the maximum operating distance - see also 5.2.1.1 d). The test shall be performed with the incandescent light source of 5.4.6.2 at the maximum intensity level at which the AOPDDR remains in normal operation. This intensity level shall be a minimum of 1 500 lx. When the AOPDDR remains in normal operation above 3 000 lx the test level shall be 3 000 lx. The measurement of the intensity of reflected light shall be carried out in the detection zone on the position of the test piece.	Tested	P

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	Both tests on the influence of incandescent light on the integrity of the detection capability shall be performed under the following conditions:		P
	- the light shall be generated by the incandescent light source as described in 5.4.6.2;		P
	- the light source shall be located outside the detection zone and the tolerance zone.		P
	The coefficient of reflection of the background used for this test shall not vary in the range of wavelengths used by the AOPDDR itself and used for the measurement of intensity.		P
	For AOPDDR-2D, the set-up has to be adapted accordingly - see 5.2.1.1 c).		P
	NOTE 1 Figure 9 shows a possible configuration for a test according to 5.2.1.2.3.		-
	NOTE 2 Figure 9 shows no detection zone because in this example it is the influence on the position accuracy that is being tested.		-
5.2.1.3	Endurance test of the detection capability		-
	It shall be verified that the detection capability is maintained by carrying out an endurance test as follows. The results of the analysis and testing according to 5.2.1.2 shall be used to determine the worst-case conditions and the appropriate test piece (see 4.2.13) to use for this test.		P
	A limited functional test B (B test) in accordance with 5.2.3.3 of IEC 61496-1:2012 shall be carried out with the ESPE in continuous operation under the worst-case conditions determined. The test piece shall be placed in a worst-case position and left in this position for a time period of 150 h.		P
	If there is more than one worst-case position, the test shall be carried out for each position of the test piece while evenly dividing the total test time of 150 h among the different worst-case positions. The possibility of zone(s) with limited detection capability shall be taken into account.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	Changes may be made to both hardware and software (if applicable) to simulate worst-case conditions.		P
	For AOPDDR-2D, the set-up shall be adapted accordingly - see 5.2.1.1 c).		P
	NOTE 1 AOPDDR with, for example, maximum undetected homogeneous and spot-like pollution on the optical window and maximum degeneration by ageing of components, etc.		-
	NOTE 2 The black test piece will lead to a lower signal-noise-ratio (S/N) than the white test piece.		-
	NOTE 3 Background with worst-case reflectance (if background interferes with measurement values).		-
	For AOPDDR-2D, the set-up shall be adapted accordingly - see 5.2.1.1 c).		P
	NOTE For AOPDDR-2D, the test piece is placed inside the detection zone(s) by half of the minimum detectable object size, i.e. the centre line of the test piece is placed at the outer border of the detection zone. For AOPDDR- 3D, the test piece is placed inside the detection zone(s) as far as the minimum detectable object size.		-
5.2.3	Limited functional tests		-
5.2.3.1	General		-
	5.2.3.1 of IEC 61496-1:2012 applies, except as follows.		P
	Addition:		-
	Unless otherwise stated in this document, either of the test pieces according to 4.2.13.2 or 4.2.13.3 shall be used for the limited functional tests.		P
	Addition:		-
5.2.9	Test pieces for type testing		-
	The stated reflectivity values of the test pieces shall be verified by inspection of the supplier's declaration (based on test results) or by measurement. Other test pieces may be used, providing they meet the relevant requirements of this document.		P
5.2.10	Position accuracy		-

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Clause	Requirement + Test	Result - Remark	Verdict
	The supplier's calculations for determining the position accuracy and the tolerance zone shall be verified for correctness and validity by comparison with the results of the measurements of the detection capability according to 5.2.1.2.		P
5.2.11	Influences on detection		-
5.2.11.1	General		-
	The requirements of 4.2.12.2 shall be verified by analysis and/or measurement. The supplier's documentation related to the reflectivity of the test piece(s) and the calculations used may be verified by inspection.		P
5.2.11.2	AOPDDR used as a trip device with orthogonal approach		-
	It shall be verified that the sensing device is actuated and the OSSDs go to and remain in the OFF-state when a test piece as defined in 4.2.13.3 is moving through the detection zone such that the direction of movement and the axis of the cylinder are orthogonal to the detection zone at a speed of 1,6 m/s. Where the supplier states that an AOPDDR can be used to detect objects moving at speeds greater than 1,6 m/s, verification shall be carried out at the stated maximum speed.		P
5.2.12	Wavelength		-
	The emitted wavelength shall be verified either by inspection of the device data sheet or by measurement.	940nm	P
5.2.13	Radiation intensity		-
	The radiation intensity shall be verified by measurement in accordance to IEC 60825-1 or IEC 62471 and by inspection of the technical documentation provided by the supplier. If the emitting element(s) uses laser technology the marking as a class 1 or class 1M laser shall be verified for correctness.		P
5.2.14	Mechanical construction		-

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Clause	Requirement + Test	Result - Remark	Verdict
	The requirements of 4.2.16 shall be verified by inspection.	Tested	P
5.3	Performance testing under fault conditions		-
5.3.2	Type 1 ESPE		-
	5.3.2 of IEC 61496-1:2012 does not apply.	Type 1 ESPE	P
5.3.3	Type 2 ESPE		-
	5.3.3 of IEC 61496-1:2012 applies, except as follows	Type 1 ESPE	N/A
	Addition:		
	It shall be verified that the drift or ageing of components that influence the detection capability will lead to an OFF-state of the OSSD(s) according to 4.3.10.		N/A
	A combination of one or more of the following is sufficient to verify the requirement to combine single faults with operating conditions/influences as required by 4.2.2.3:		N/A
	- analysis;		N/A
	- simulation;		N/A
	- tests carried out in the presence of a single fault, where relevant.		N/A
5.3.4	Type 3 ESPE		-
	5.3.4 of IEC 61496-1:2012 applies, except as follows.	Type 1 ESPE	N/A
	Addition:		-
	It shall be verified that the drift or ageing of components that influence the detection capability will lead to an OFF-state of the OSSDs within a time period of 5 s according to 4.3.10.		N/A
	A combination of one or more of the following is sufficient to verify the requirement to combine single faults with operating conditions/ influences as required by 4.2.2.4:		N/A
	- analysis;		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	- simulation;		N/A
	- tests carried out in the presence of a single fault, where relevant.		N/A
5.3.5	Type 4 ESPE		-
	5.3.5 of IEC 61496-1:2012 does not apply.	Type 1 ESPE	N/A
5.4	Environmental tests		-
5.4.2	Ambient temperature variation and humidity		--
	5.4.2 of IEC 61496-1:2012 applies, except as follows.		P
	Addition:		-
	The ESPE shall be subjected to the following condensing test:		-
	- the ESPE shall be supplied with its rated voltage and stored in a test chamber at an ambient temperature of 5 °C for 1 h;		P
	- the ambient temperature and the humidity shall be changed within a time period of up to 2 min to a temperature of (25 ± 5) °C and a relative humidity of (70 ± 5) %;		P
	- a C test shall be performed with a duration of 10 min using the black test piece (see 4.2.13.2);		P
	- if a restart interlock is available it shall not be operational during the C test;		P
	- to verify the stated detection capability of the ESPE during the C test, either		P
	a) the ESPE shall be operated with a detection zone set up as described in 5.1.1.2 and a distance between the origin of distance measurement and the test piece of BTD, or		P
	b) measurement values shall be used for verification.		P
5.4.4	Mechanical influences		-
5.4.4.1	Vibration		-
	Replacement:		-

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Clause	Requirement + Test	Result - Remark	Verdict
	The test sample shall be exposed to vibration tests according to IEC 60068-2-6. The test shall be carried out without anti-vibration mountings.		P
	NOTE 1 The tests are intended to demonstrate a minimum robustness of the device under test.		-
	The following conditions shall apply:		P
	- Frequency range: 10 Hz to 55 Hz;		P
	- Sweep rate: 1 octave/min;		P
	- Amplitude: 0,35 mm ± 0,05 mm;		P
	- Number of sweeps: 20 for each of three mutually perpendicular axes (no delay at resonant frequencies).		P
	The following limited functional tests shall be performed for each axis:		P
	- an A test shall be performed during each of the first and last sweeps;		P
	- a B test shall be carried out, so that the test piece is brought into the detection zone at the beginning of the second sweep and is removed at the end of the 19th sweep.		P
	If the supplier specifies the AOPDDR for use on ground vehicles the supplier shall choose an appropriate 5M class according to IEC TR 60721-4-5. The related tests shall be carried out as C tests.		P
	NOTE 2 Example for ground vehicles include forklift trucks, automated guided vehicles (see also IEC 60721-3-5).		-
	If the AOPDDR is not intended to be mounted on a machine (i.e. not intended to be subjected to high vibration), the levels of amplitude and frequency may be reduced for the A test depending on the intended application. In this case, a C test may be carried out instead of the B test.		P
	At the end of the tests, the AOPDDR shall be inspected for the absence of damage including displacement and/or cracks of the optical window. It shall be verified by test that the detection zone has not changed in its orientation, size or position.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	Replacement, including the title of 5.4.4.2 of IEC 61496-1:2012:		P
5.4.4.2	Shock		-
	The test sample shall be exposed to shock tests according to IEC 60068-2-27.	Tested	P
	The following conditions shall apply:		-
	- Acceleration: 100 m/s^2;		P
	- Duration of pulse: 16 ms;		P
	- Number of shocks/direction: 1 000 ± 10;		P
	- Directions of shocks: 6 with 3 mutually perpendicular axes.		P
	The following tests shall be performed for each direction:		P
	- an A test shall be performed during each of the first and last (100 ± 10) shocks;		P
	- a B test shall be carried out, so that the test piece is brought into the detection zone after the first (100 ± 10) shocks.		P
	If the supplier specifies the AOPDDR for use on ground vehicles the supplier shall choose an appropriate 5M class according to IEC TR 60721-4-5. The related tests shall be carried out as C tests.		P
	NOTE Example for ground vehicles include forklift trucks, automated guided vehicles (see also IEC 60721-3-5).		-
	If the AOPDDR is not intended to be mounted on a machine (i.e. not intended to be subjected to shock), the test conditions may be reduced for the A test depending on the intended application. In this case, a C test may be carried out instead of the B test.		P
	At the end of the tests the AOPDDR shall be inspected for the absence of damage including displacement and/or cracks of the optical window. It shall be verified by test that the detection zone has not changed in its orientation, size or position.		P
	Addition:		-

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Clause	Requirement + Test	Result - Remark	Verdict
5.4.4.3	Change of temperature		-
	The ESPE shall be subjected to a test Na according to IEC 60068-2-14 using the following relevant values and conditions:	Tested	P
	- low temperature TA: - 25 °C;		P
	- high temperature TB: 70 °C;		P
	- four cycles;		P
	- ESPE not energized during the temperature cycles;		P
	- duration t_1: 60 min;		P
	- following the test, the AOPDDR shall be inspected for absence of damage including displacement and/or cracks of the optical window;	No damage after test	P
	- a B test shall be carried out in the test environment according to 5.1.2.1 of IEC 61496-1:2012 to verify that the ESPE is capable of continuing in normal operation.		P
5.4.4.4	Hammer tests		-
5.4.4.4.1	General		-
	The ESPE shall be subjected to tests according to IEC 60068-2-75 using the following values and conditions:	Tested	P
	- three impacts;		P
	- mounting by its normal means on a rigid support;		P
	- no initial measurements;		P
	- the impacts shall be directed at the centre of the optical window in the plane of the detection zone for AOPDDR-2D and the centre axis for AOPDDR-3D;		P
	- ESPE not energized during the impacts.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	The test of 5.4.4.4.2 shall be carried out after the change-of-temperature test of 5.4.4.3 has been completed and before the test of 5.4.5. The test of 5.4.4.4.3 shall be carried out after the test of 5.4.5 has been completed. All these tests shall be carried out with the same device sample.		P
5.4.4.4.2	Normal operation		-
	To test that the ESPE is capable of continuing in normal operation after the impacts according to IEC 60068-2-75, the following values and conditions shall be used:		P
	- impact energy of 0,5 J;		P
	- following the test, the AOPDDR shall be inspected and shall not have any displacement or cracks of the optical window;		P
	- a B test shall be carried out placing the test piece at each position where the stated detection capability might be reduced by the impacts.		P
5.4.4.4.3	Fail to danger		-
	To test that the ESPE will not fail to danger after the impacts according to IEC 60068-2-75, the following values and conditions shall be used:		P
	- impact energy of 2,0 J;		P
	- following the test, the AOPDDR shall be inspected for displacement and/or cracks of the optical window;		P
	- a C test shall be carried out placing the test piece at each position where the stated detection capability might be reduced by the impacts.		P
5.4.5	Enclosures		-
	Replacement:		-
	The requirements of 4.3.4 of this document for degrees of protection shall be tested in accordance with IEC 60529 after the tests of 5.4.4 (excluding 5.4.4.4.3) have been completed. The remaining requirements shall be verified by inspection.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	Additional subclauses:		-
5.4.6	Light interference on AOPDDR receiving elements and other optical components		-
5.4.6.1	General		-
	All tests shall be carried out with the black test piece (see 4.2.13.2). During the B tests and C tests, the test piece shall be introduced into the detection zone in such a manner that the interfering light is not interrupted. The test piece shall first be positioned near the centre axis and then at one of the corner axes at a uniform distance from the AOPDDR for the test.		P
	The tests described in 5.4.6.4.3, 5.4.6.4.4, 5.4.6.5.4, 5.4.6.5.5, 5.4.6.6.3 and 5.4.6.7.3 shall only be carried out if the AOPDDR contains optical components, other than those necessary for the sensing function or measurement of distance, which may be influenced by interfering light. Analysis of the characteristics and the intended function of the other optical components shall be carried out to determine if additions to, or combinations of, test conditions are required in order to detect possible failure to danger of the ESPE (for example, to verify the absence of failure to danger of the ESPE due to pollution monitoring means in the presence of light interference).		P
	NOTE Other optical components can include the following: emitting/receiving elements, reflectors, lenses, etc., provided within the AOPDDR.		-
	Table 2 gives an overview of the light interference tests.		P
5.4.6.2	Light sources		-
	The light sources shall be as follows.		-
	a) Incandescent light source: a linear tungsten halogen (quartz) lamp with the following characteristics:		P
	- colour temperature: 2 800 K to 3 200 K;		P

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Clause	Requirement + Test	Result - Remark	Verdict
	- rated voltage: any value within the range 100 V to 250 V;		P
	- supply voltage: rated voltage ± 5 %, sinusoidal AC at 48 Hz to 62 Hz;		P
	The lamp shall be mounted in a parabolic reflector of minimum dimensions 200 mm × 150 mm, having a diffuse reflective surface and a reflectivity that is uniform within ±5 % over the wavelength range 400 nm to 1 500 nm.		P
	NOTE This source produces a beam of near-uniform intensity with known spectral distribution and having a predictable modulation at twice the supply frequency.		-
	b) Fluorescent light source: a linear fluorescent tube with the following characteristics:		P
	- size: T8 × 600 mm minimum;		P
	- colour temperature: 3 000 K to 6 500 K;		P
	The linear fluorescent tube shall be used in combination with an electronic ballast having the following characteristics:		P
	- operating frequency: 25 kHz to 50 kHz;		P
	- power rating corresponding to the tube;		P
	The linear fluorescent tube shall be operated at its rated power supply voltage ± 5 %, without a reflector or diffuser.		P
	Other fluorescent light sources having, for example, electronic ballasts with an operating frequency other than that specified can lead to different test results. Therefore, the use of other types of fluorescent light sources or a light source generator simulating the effects of different fluorescent light sources should be considered for testing.		P
	c) Flashing beacon light source: a light source employing a xenon flash tube (without enclosure, reflector or filter) having the following characteristics:		P
	- flash duration: from 40 µs to 1 200 µs (measured to the half-intensity point);		P

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Clause	Requirement + Test	Result - Remark	Verdict
	- flash frequency: 0,5 Hz to 2 Hz;		P
	- input energy per flash: 3 J to 5 J;		P
	d) Laser beam pointer: a collimated laser beam having characteristics within the following limits:		P
	- flash duration: continuous wave mode;		P
	- wavelength: within 550 nm up to 670 nm;		P
	- beam shape: diameter below 5 mm;		P
	- light intensity 0,7 mW up to 1 mW;		P
	- laser Class 2.		P
	WARNING - The laser beam pointer is a class 2 device. Never look into the beam.		-
5.4.6.3	Test sequences		-
	1) Test sequence 1: a) OSSDs of the ESPE in on-state. b) Switch on interfering light (the OSSDs shall remain in on-state). c) B test. d) Switch off ESPE for 5 s. Restore power. Reset start interlock if fitted. e) B test. f) Switch off interfering light. g) B test.	In the test, monitoring the output distance instead of the status of the OSSD.	P
	2) Test sequence 2: a) OSSDs of the ESPE in on-state. b) Switch on interfering light. c) C tests repetitively for 1 min. d) Switch off the ESPE for 5 s. Restore power. Reset start interlock if fitted. e) C tests repetitively for 1 min. f) Switch off interfering light. g) C tests repetitively for 1 min.	In the test, monitoring the output distance instead of the status of the OSSD.	P

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Clause	Requirement + Test	Result - Remark	Verdict
	3) Test sequence 3: a) OSSDs of the ESPE in on-state. b) Switch on interfering light. c) C tests repetitively for 3 min.	In the test, monitoring the output distance instead of the status of the OSSD.	P
5.4.6.4	Light interference - Incandescent light		-
5.4.6.4.1	Normal operation - Interference on AOPDDR receiving elements		-
	The ESPE shall be subjected to a test using test sequence 1 of 5.4.6.3 with the incandescent light source of 5.4.6.2 producing a light intensity of 1 500 lx ± 10 % measured in the position of the housing of the AOPDDR. The light shall be directed along the centre axis of an AOPDDR and the light source shall be located outside the detection zone and the tolerance zone. The ESPE shall not go to the ON-state when the test sequence requires it to be in the OFF-state. If the ESPE goes to the OFF-state when the test sequence requires it to be in the ON-state, the following additional tests a) and b) shall be performed.		P
	NOTE The light source used for test has typically a distance of 1,6 m to 2,2 m from the housing and will be located outside the detection zone and the tolerance zone for the test as described in a) and b).		-

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Clause	Requirement + Test	Result - Remark	Verdict
	a) If the minimum detection zone or tolerance zone is so large, that the light source is inside the detection zone or the tolerance zone producing a light intensity of 1 500 lx ± 10 % for the test, the light source shall be outside the detection zone and the tolerance zone during the test, but close to the border of the tolerance zone. The ESPE shall continue in normal operation during the test sequence 1 of 5.4.6.3, using the incandescent light source of 5.4.6.2. The light source shall be located as close as possible to the plane of the detection zone for an AOPDDR-2D. For an AOPDDR-3D the light source shall be located adjacent to the corner axis. The light source shall not be detected by the ESPE during the test and the distance between the ESPE and the light source shall be the minimum distance at which the ESPE is able to pass an A test. If the intensity measured in front of the AOPDDR receiver is less than 1 500 lx, then the accompanying documents shall contain instructions regarding the avoidance of interference by incandescent light sources (see Clause 7, item ppp)).		P
	b) The ESPE shall continue in normal operation during the test sequence 1 of 5.4.6.3 using the incandescent light source of 5.4.6.2. The light source shall be located in the plane of the detection zone for an AOPDDR-2D. For an AOPDDR-3D the light source shall be located in the centre axis for an AOPDDR-3D. The distance between the ESPE and the light source shall be the minimum distance at which the ESPE is able to pass an A test. If the intensity measured in front of the AOPDDR receiver is less than 1 500 lx, then the accompanying documents shall contain instructions regarding the avoidance of interference by incandescent light sources (see Clause 7, item ppp)).		P
5.4.6.4.2	Failure to danger - Interference on AOPDDR receiving elements		-

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Clause	Requirement + Test	Result - Remark	Verdict
	There shall be no failure to danger of the ESPE during test sequence 2 of 5.4.6.3 using the incandescent light source of 5.4.6.2 producing a light intensity of 3 000 lx ± 10 % measured in the position of the housing of the AOPDDR. The light shall be directed along the centre axis of an AOPDDR. If the light source is inside the detection zone or tolerance zone for this test, the following additional tests a) and b) shall be performed instead of the test described above in this paragraph.		P
	a) There shall be no failure to danger of the ESPE during test sequence 2 of 5.4.6.3 using the incandescent light source of 5.4.6.2 producing a light intensity of 3 000 lx ± 10 %. The light source shall be located as close as possible to the plane of the detection zone for an AOPDDR-2D. For an AOPDDR-3D the light source shall be located adjacent to the corner axis. The light source shall not be detected by the ESPE during the test.		P
	b) There shall be no failure to danger of the ESPE during test sequence 2 of 5.4.6.3 using the incandescent light source of 5.4.6.2. The light source shall be placed in the plane of the detection zone for an AOPDDR-2D. For an AOPDDR-3D the light source shall be located in the centre axis. The light source shall be outside the detection zone and the tolerance zone during the test, but close to the border of the tolerance zone. The C tests shall be carried out with the axis of the test piece placed on the furthest boundary of the detection zone.		P
5.4.6.4.3	Normal operation - Interference on other optical components		-
	The ESPE shall continue in normal operation during test sequence 1 of 5.4.6.3 using the incandescent light source of 5.4.6.2 producing a light intensity of 1 500 lx ± 10 %. The incandescent light source shall be placed in the position(s) that allows normal operation and where other optical components can be influenced.		N/A
5.4.6.4.4	Failure to danger - Interference on other optical components		-

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Clause	Requirement + Test	Result - Remark	Verdict
	There shall be no failure to danger of the ESPE during test sequence 2 of 5.4.6.3 using the incandescent light source of 5.4.6.2 producing a light intensity of 3 000 lx ± 10 %. The incandescent light source shall be placed in the position(s) where other optical components can be influenced.		N/A
5.4.6.5	Light interference - Fluorescent light		-
5.4.6.5.1	General		-
	This test shall be performed with three variations, using light from the centre and light from each end (anode and cathode areas) of the tube.		P
	NOTE One aim of the test using the fluorescent light source is to check the susceptibility of the AOPDDR to high frequency optical radiation.		-
5.4.6.5.2	Normal operation - Interference on AOPDDR receiving elements		P
	The test shall be carried out with the minimum detection zone possible, but the operating distance plus tolerance zone shall be ⩾0,2 m. The ESPE shall continue in normal operation during test sequence 1 of 5.4.6.3 using the fluorescent light source of 5.4.6.2 placed outside the detection zone and the tolerance zone, but close to the border of the tolerance zone in the plane of the detection zone for an AOPDDR-2D. For an AOPDDR-3D the light source shall be placed in the centre axis.		P
5.4.6.5.3	Failure to danger - Interference on AOPDDR receiving elements		-
	The test shall be carried out with the maximum detection zone possible. There shall be no failure to danger of the ESPE during test sequence 2 of 5.4.6.3 using the fluorescent light source of 5.4.6.2 placed at a distance of 0,2 m from the housing of the AOPDDR in the plane of the detection zone for an AOPDDR-2D. For an AOPDDR-3D the light source shall be placed in the centre axis. The C tests shall be carried out with the axis of the test piece placed on the furthest boundary of the detection zone.		P
	NOTE The lamp body can be detected as an object during this test.		-

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Clause	Requirement + Test	Result - Remark	Verdict
5.4.6.5.4	Normal operation - Interference on other optical components		-
	The test shall be carried out with the minimum detection zone possible, but the operating distance plus tolerance zone shall be ⩾ 0,2 m. The ESPE shall continue in normal operation during test sequence 1 of 5.4.6.3 using the fluorescent light source of 5.4.6.2 placed at a distance of 0,2 m from the housing of the AOPDDR in the position(s) where other optical components can be influenced by light interference. If this position(s) coincides with, or meets, the detection zone of the AOPDDR, the fluorescent light source shall be placed as close as possible but ⩾ 0,2 m, so that the body of the lamp is not detected.		N/A
5.4.6.5.5	Failure to danger - Interference on other optical components		-
	The test shall be carried out with the maximum detection zone. There shall be no failure to danger of the ESPE during test sequence 2 of 5.4.6.3 using the fluorescent light source of 5.4.6.2 placed at a distance of 0,2 m to the housing of the AOPDDR in the position(s) where other optical components can be influenced by light interference. The C tests shall be carried out with the axis of the test piece placed on the furthest boundary of the detection zone.		N/A
	NOTE The lamp body can be detected as an object during this test.		-
5.4.6.6	Light interference - Flashing beacon		-
5.4.6.6.1	General		-
	The position of the flashing beacon shall be fixed during the tests. The required C tests shall be carried out with the axis of the test piece placed on the furthest boundary of the detection zone for an AOPDDR-2D. The required C tests shall be carried out with the test piece placed in a position that is both inside and adjacent to the furthest boundary of the detection zone for an AOPDDR-3D.		P

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Clause	Requirement + Test	Result - Remark	Verdict
5.4.6.6.2	Failure to danger - Interference on AOPDDR receiving elements		-
	There shall be no failure to danger of the ESPE during test sequence 3 of 5.4.6.3 using the flashing beacon of 5.4.6.2 placed at a distance of 3,0 m from the housing of the AOPDDR in the plane of the detection zone for an AOPDDR-2D. For an AOPDDR-3D the light source shall be placed in the centre axis.		P
5.4.6.6.3	Failure to danger - Interference on other optical components		-
	There shall be no failure to danger of the ESPE during test sequence 3 of 5.4.6.3 using the flashing beacon of 5.4.6.2 placed at a distance of 3,0 m from the housing of the AOPDDR in the position(s) where other optical components can be influenced by light interference.		N/A
5.4.6.7	Light interference - Laser beam pointer		-
5.4.6.7.1	General		-
	The position of the laser beam pointer shall be fixed during the tests. The required C tests shall be carried out with the axis of the test piece placed at one position on the furthest boundary of the detection zone for an AOPDDR-2D. The required C tests shall be carried out with the test piece placed in a position that is both inside and adjacent to the furthest boundary of the detection zone for an AOPDDR-3D.	Tested	P
5.4.6.7.2	Failure to danger - Interference on AOPDDR receiving elements		-
	There shall be no failure to danger of the ESPE during test sequence 3 of 5.4.6.3 using the laser beam pointer of 5.4.6.2 placed at a distance of 3,0 m from the housing of the AOPDDR in the plane of the detection zone for an AOPDDR-2D. For an AOPDDR-3D the light source shall be placed in the centre axis.		P
5.4.6.7.3	Failure to danger - Interference on other optical components		-

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Clause	Requirement + Test	Result - Remark	Verdict
	There shall be no failure to danger of the ESPE during test sequence 3 of 5.4.6.3 using the laser beam pointer of 5.4.6.2 placed at a distance of 3,0 m from the housing of the AOPDDR in the position(s) where other optical components can be influenced by light interference.		N/A
5.4.6.8	Light interference by an emitting element of identical design	See the manual	-
5.4.6.8.1	General		-
	In order to test for interference between AOPDDRs of identical design, two devices shall be mounted in a position and angle representative of the worst-case conditions as determined by analysis. A possible configuration for this test of AOPDDR-3D is shown in Figure 12 and Figure 13.		N/A
	NOTE 1 For the test of 5.4.6.8.3, the worst-case conditions for this test can include maximum detection zones, opposite mounting orientation of the AOPDDRs and positioning the test piece just beside the centre axes as shown in Figure 12.		-
	For the tests of 5.4.6.8.2 and 5.4.6.8.3, an exact positioning of the devices under test is required in such a way that the emitting element beam(s) of one AOPDDR is (are) directed exactly to the receiving element(s) of the other AOPDDR.		N/A
	NOTE 2 An infrared camera can be used for exact positioning.		-
	For AOPDDR-2D the set-up has to be adapted accordingly - see 5.2.1.1 c).		N/A
	NOTE Figure 12 shows a possible configuration for the tests of 5.4.6.8.2 (without test piece) and 5.4.6.8.3.		-
	NOTE The minimum dimension of the overlap of “10” and “11” is 2 times the tolerance zone plus two times the minimum detectable object size.		-
	The reflective background shall have a coefficient of diffuse reflection greater or equal 80 % at the emitted wavelength (named white background). The white test piece (see 4.2.13.3) shall be used for this test.		N/A
	The distance between the background and the nearest part of the tolerance zone should be chosen according to analysis.		N/A

SGS-CSTC Standards Technical Services (Shanghai) Co.,Ltd. EEC Safety Lab

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EN IEC 61496-3			
Clause	Requirement + Test	Result - Remark	Verdict
5.4.6.8.2	Normal operation		-
	The information for use may contain instructions regarding the avoidance of interference between two or more AOPDDRs of identical design (for example, by special mounting). If no mounting restrictions are given by the supplier for the AOPDDR, an A test shall be carried out with both AOPDDRs for a time period of 4 h when radiation from the emitting element(s) of an AOPDDR of identical design is directed towards the receiving element(s) of the other AOPDDR according to Figure 12, and Figure 13 without the test piece.		N/A
5.4.6.8.3	Failure to danger		-
	There shall be no failure to danger of the ESPE when radiation from the emitting element(s) of an AOPDDR of identical design is directed towards the receiving element(s) of the other AOPDDR according to Figure 12 and Figure 13. This test shall be carried out for both AOPDDRs for a time period of 4 h. None of the devices under test shall go to the ON-state.		N/A
5.4.7	Pollution interference		-
5.4.7.1	General		-
	Immunity against pollution interference shall be tested by carrying out tests simulating spotlike pollution and homogeneous pollution. The tests listed in 5.4.7.2 and 5.4.7.3 may not be sufficient to cover all possible designs of pollution monitoring means. In such cases, additional analysis and tests shall be carried out to verify the stated detection capability. As an example, it may be required to consider the variation of the reflectivity of a reference object or the transmission capability of optical components. Special attention shall be paid to the influence of temperature on the pollution monitoring means.		P
5.4.7.2	Pollution test with opaque test spot		-
	Immunity against spot-like pollution shall be tested as follows.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	- Spot-like pollution shall be simulated by using circular opaque test spots of three different diameters:		P
	• half of one emitter aperture (average) at the optical window, but resulting in a spot diameter not exceeding 20 mm;		P
	• half of one receiver aperture (average) at the optical window, but resulting in a spot diameter not exceeding 20 mm;		P
	• 10 mm		P
	- The coefficient of diffuse reflection of the test spots at the emitted wavelength shall be within the range of 18 % to 22 %.		P
	- During the test the spots shall be placed individually at all position relevant to the detection capability of the AOPDDR.		P
	- Test whether the simulated spot-like pollution will lead to an OFF-state of the OSSDs within a time period of 5 s or does not reduce the stated detection capability.		P
	- Tests shall be carried out to verify that when simulated pollution leads to an OFF-state of the OSSDs, actuation of the restart interlock (if applicable) or a new power-up does not lead to an ON-state of the OSSDs. If a restart interlock is fitted, the OSSDs shall stay in the OFF-state when the simulated pollution is removed.		P
	NOTE 1 For the purposes of this document, the diameter of a Gaussian laser beam is defined by the 1/e2 intensity levels.		-
	Based on the analysis required in 5.4.7.1, the diameter of the test spot may be adapted to more critical cases (e.g. diameter a twice - see Figure 14 and Figure 15).		P
	NOTE 2 The emitting element of an AOPDDR can consist of one or more LEDs. Half of one emitter aperture is the half diameter of the aperture a of one LED projected to the plane of the optical window (see Figure 14).		-

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Clause	Requirement + Test	Result - Remark	Verdict
	NOTE 3 The receiving element of an AOPDDR can consist out of a pixel array with one or more lenses in front of the pixel array. The half of one receiver aperture is the half diameter of the aperture of one pixel projected to the plane of the optical window (see Figure 15).		-
5.4.7.3	Test of homogeneous pollution of the emitter and receiver aperture		P
	Immunity against homogeneous pollution shall be tested as follows.		P
	- Homogeneous pollution shall be simulated by using a grey, half-tone foil with a line frequency greater than four lines per millimetre. Reflections produced by such foil shall not influence the test results.		P
	- For an AOPDDR having a curved optical window, the foil shall cover a 45° arc over the emitter and receiver aperture area(s) of the optical window of the housing. For an AOPDDR having an optical window with a flat characteristic, the foil shall cover 25 % of the emitter and receiver aperture area(s) of the optical window of the housing, but in any case the foil(s) shall have a minimum size of 300 mm2.		P
	- During the test the foil shall be placed at all positions within the emitter and receiver aperture area(s) at the optical window relevant to the detection capability of the AOPDDR. Based on analysis, this test can require positioning either separately on emitter or receiver or require covering a portion of both.		P
	It can be necessary for the test to use more than one foil.		P
	- Test whether simulated homogeneous pollution outside the limits specified by the supplier will lead to an OFF-state of the OSSDs within a time period of 5 s.		P
	- Test whether the AOPDDR continues in normal operation when the received signal energy of the detection system is attenuated up to 30 % by simulated homogeneous pollution.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	- Tests shall be carried out to verify that when simulated pollution leads to an OFF-state of the OSSDs, actuation of the restart interlock (if applicable) or a new power-up does not lead to an ON-state of the OSSDs. If a restart interlock is fitted, the OSSDs shall stay in the OFF-state when the simulated pollution is removed.		P
	NOTE 1 Equivalent material for the simulation of homogeneous pollution, for example powder or density filters, can be used.		-
	NOTE 2 In certain applications, for example, in a dusty environment, the rate of accumulation of pollution on the optical window of the AOPDDR can be influenced by the mounting position and orientation of the AOPDDR.		-
5.4.8	Interference by surrounding objects		-
5.4.8.1	Background interference		-
	If the position accuracy within the detection zone can be influenced by the background, the supplier shall identify the worst-case conditions regarding background interference.		P
	The test covering background interference on the detection capability shall be carried out according to 5.2.1.2 and Table 1, using the following background:		P
	a) a corner cube reflector with a coefficient of reflection $\geqslant$ 330 cd • lx^{-1} • m^{-2};		P
	b) a diffuse reflector with a coefficient of diffuse reflection of 1,8 % to 5 %;		P
	c) other relevant background material having a reflectivity between a) and b), if such a background is expected to have a greater influence on the detection capability.		P
	The worst-case distance between test piece and background shall be determined by measurement.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	If the supplier specifies the maximum reflectivity that is monitored by the AOPDDR, a test shall be carried out to verify that the reflectivity of a background that exceeds the specified maximum reflectivity leads to an OFF-state of the OSSDs within the specified response time. In this case, the background interference test according to a) above shall be carried out with the specified maximum reflectivity with the OSSDs remaining in the ON-state when the detection zone is not penetrated, instead of testing with a corner cube reflector.		P
	Figure 16 shows a possible configuration for the tests of 5.4.8.1.		P
5.4.8.2	Multi-path reflections		-
	Analysis shall be conducted to determine if the position accuracy can be influenced by multipath reflections. The supplier shall identify appropriate tests based on the test set up given at Figures 17 and 18 but using worst-case considerations regarding multi-path reflections. Worst case conditions should take foreseeable industrial environments into account. They can be represented by variations in reflectivity, distances and angles, for example, highly reflective surfaces in the nearby background. As long as analysis does not show the necessity of other tests the test setups shown in Figure 17 (top view) and Figure 18 (side view) shall be used.		N/A
	NOTE Highly reflective surfaces that can interfere with the measurement results include material that generate specular reflections.		-
	If the position accuracy can be influenced by multi-path reflections, the supplier shall identify appropriate tests taking worst-case considerations regarding multi-path reflections into account.		N/A
	The test setups shall be as follows:		N/A
	- the target orientation is orthogonal to the centre axis in one direction and 45 ° tilted in the other direction;		N/A
	- the target size is 0,5 m × 0,5 m;		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	- the target shall have a coefficient of diffuse reflection greater or equal 80 % at the emitted wavelength;		N/A
	- the distance between the origin (of distance measurement) and the target midpoint shall be the minimum operating distance (but 0,2 m as a minimum) and the maximum operating distance;		N/A
	- the reflective background size of 1,0 m height and 2,0 m width;		N/A
	- the reflective background shall have a coefficient of diffuse reflection greater or equal 80 % at the emitted wavelength (named white background) and lower or equal 10 % at the emitted wavelength (named black background);		N/A
	- the reflective background shall be positioned for the tests as shown in Figures 17 and 18;		N/A
	- the distances between the AOPDDR centre axis and the reflective background shall be 0,25 m and 1,0 m;		N/A
	- the surrounding environment should be considered.		N/A
	The test setups shall be combined for test as follows, and test shall be carried out at minimum operating distance (but 0,2 m as a minimum) and the maximum operating distance:		N/A
	- Test A: black background with 0,25 m distance between the AOPDDR centre axis and black background;		N/A
	- Test B: white background with 0,25 m distance between the AOPDDR centre axis and white background;		N/A
	- Test C: black background with 1,0 m distance between the AOPDDR centre axis and black background		N/A
	- Test D: white background with 1,0 m distance between the AOPDDR centre axis and white background;		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	The tests shall be carried out by acquiring distance measurement values from the AOPDDR for the tests described above to determine the maximum difference of measurement values caused by multi-path reflections. A minimum of 1 000 measurements shall be used for determining position. Measurement values shall be taken at the target centre taking into account a square area with each side corresponding to the stated detection capability.		N/A
	The results of this test shall be taken into account to determine the tolerance zone in accordance with 4.2.12.2.		N/A
	The resulting differences of average value and standard deviation σ of the test setups A and B respectively C and D shall be considered in the determination of the tolerance zone.		N/A
5.4.9	Manual interference		-
5.4.9.1	Tests with opaque test spots		-
	Immunity against manual interference shall be tested as follows.	Tested	P
	- Spot-like manual interference shall be simulated by using two circular opaque test spots of 15 mm diameter. The first shall have a coefficient of diffuse reflection from 18 % to 22 % at the emitted wavelength. The second shall be a corner cube reflector with a coefficient of reflection ⩾ 330 cd • lx^{-1} • m^{-2}.		P
	- During both tests the spots shall be placed on the optical window and, where appropriate, within the zone(s) with limited detection capability at any position relevant to the detection capability of the AOPDDR (see 4.1.6).		P
	- Tests shall be carried out to verify that simulated manual interference either leads to an OFF-state of the OSSDs within a time period of 5 s or the stated response time for AOPDDR for use as trip device with orthogonal approach or does not reduce the stated detection capability.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	- Tests shall be carried out to verify that when simulated manual interference leads to an OFF-state of the OSSDs, actuation of the restart interlock (if applicable) or a new powerup does not lead to an ON-state of the OSSDs. If a restart interlock is fitted, the OSSDs shall stay in the OFF-state when the simulated manual interference is removed.		P
	The second test shall be carried out with a test spot of lower reflectivity if the device is designed as described in the paragraph before the last in 5.4.8.1. The test spot shall have the maximum reflectivity with the AOPDDR remaining in normal operation.		P
	NOTE 1 These tests simulate manual interference by small objects such as adhesive tape or cigarette lighters.		-
	NOTE 2 The test for pollution interference with the opaque test spot according to 5.4.7.2 also serves to test for immunity against manual interference.		-
5.4.9.2	Manual interference test with AOPDDR covered		-
	Tests for immunity against coverage shall be performed as follows.	Tested	P
	- The materials used for coverage shall have a coefficient of diffuse reflection below 10 % representing black and as defined for the white test piece and the retro-reflective test piece (see 4.2.13).		P
	- The test shall be performed using the materials defined above within the zone(s) with limited detection capability (see 4.1.6) by covering 50 % of the optical window of the housing at any relevant location, providing that at least one receiver or emitter beam is covered.		P
	The test shall be performed in the following sequence.		P
	a) AOPDDR in normal operation.		P
	b) Switch off the AOPDDR. Introduce the materials used for the coverage. Restore power. Reset start interlock if fitted.		P
	c) The OSSDs shall stay in the OFF-state at least until the coverage is removed.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	d) Remove the coverage. Reset start interlock if fitted.		P
	e) Introduce the material used for the coverage.		P
	f) Verify that the OSSDs go to the OFF-state within a time period of 5 s or the stated response time for AOPDDR for use as trip device with orthogonal approach, and remain in the OFF-state.		P
	Additional tests by covering greater angles or areas than those defined above shall be carried out if it is likely that such coverage may not be detected.		P
5.4.9.3	Manual interference test in the zone with limited detection capability		-
	Immunity against manual interference in the zone with limited detection capability shall be tested as follows.	Tested	P
	- The object used for simulating manual interference shall be a cylinder with a minimum effective length of 0,3 m. The surface of the object shall have a coefficient of diffuse reflection within the range of 80 % to 90 % at the emitted wavelength.		P
	- During the test the object shall be used orthogonal to the detection zone (AOPDDR-2D) and orthogonal to the centre axis and corner axis (AOPDDR-3D).		P
	- The diameter of the object shall be 5 mm.		P
	- The detection zone shall be set to maximum, when applicable.		P
	- The test shall be carried out by placing the object in the zone with limited detection capability.		P
	- The black test piece (see 4.2.13.2) shall be used for the C tests to be performed.		P
	- The tests shall be performed to verify that the stated detection capability is maintained in the presence of an object in the zone with limited detection capability. The black test piece shall be moved through the optical occlusion caused by the object as close as possible to the object and at the stated maximum detection distance.		P

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	This test can be omitted if appropriate measures (for example mounting restrictions) regarding the avoidance of objects with a diameter equal or greater than 5 mm in the zone with limited detection capability is provided within the information for use.	No this situation	N/A
5.4.10	Optical occlusion in the detection zone (eclipsed by small object)		-
	Immunity against optical occlusion within the detection zone shall be tested as follows.	Tested	P
	- The object used for simulating optical occlusion shall be a cylinder with a minimum effective length of 0,3 m. The surface of the test piece shall have a coefficient of diffuse reflection of less than 20 % at the emitted wavelength.		P
	- During the test the occluding object shall be used orthogonal to the detection zone (AOPDDR-2D) and orthogonal to the centre axis and corner axis (AOPDDR-3D).		P
	- The diameter of the occluding object shall be 5 mm unless determined otherwise by the analysis of 4.3.9.		P
	- The detection zone shall be set to maximum, when applicable.		P
	- The test shall be carried out by placing the occluding object in the detection zone as near as possible to the AOPDDR with the OSSDs in the ON-state.		P
	- The black test piece (see 4.2.13.2) shall be used for the B-tests to be performed.		P
	- The tests shall be performed to verify that the stated detection capability is maintained in the presence of occluding shadowing. The black test piece shall be moved through the optical occlusion of the occluding object as close as possible to the occluding object and at the stated maximum detection distance.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	- If the AOPDDR stays in the OFF-state after placing the occluding object anywhere in the detection zone the requirements of the test are fulfilled also.		P
	Additional tests shall be carried out when the analysis of 4.3.9 shows that the following can affect the immunity to optical occlusion:		P
	- distances between the AOPDDR and the shadowing object other than those stated above;		P
	- dimensions of the detection zone other than the maximum;		P
	- other distances between the occluding object and the test piece;		P
	- different diameters of the occluding object at different distances from the AOPDDR;		P
	- different positions of the occluding object in front of the AOPDDR (for example, different angles); and/or		P
	- more than one occluding object.		P
6	**Marking for identification and for safe use**		**-**
6.1	General		-
	6.1 of IEC 61496-1:2012 applies, except as follows.		P
	Addition to the second list of the subclause:		-
	l) indication of the plane of detection for an AOPDDR-2D;		P
	m) indication of the origin of distance measurement.		P
	The markings required by 6.1 b), c) and d) of IEC 61496-1:2012 and 6.1 m) may alternatively be given in the accompanying documents.		P
7	**Accompanying documents**		**-**
	Clause 7 of IEC 61496-1:2012 applies except as follows.	See the manual	P
	Addition:		-
	The accompanying documents shall contain the following information where applicable:		-

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Clause	Requirement + Test	Result - Remark	Verdict
	aaa) application examples showing the tolerance zone(s);		P
	bbb) dimensions of maximum and minimum detection zone(s) and tolerance zone(s) together with information about the origin of distance measurement (see also Figure 1 and Figure 2) for the determination of the detection range;		P
	ccc) information about the minimum required distance between the border of a detection zone and the surrounding environment without detecting, for example, walls or parts of machines in order to ensure availability in operation (see also AA.5.2);		P
	ddd) instructions for setting the detection zone(s) including consideration of the tolerance zone(s) and details on other optional functions of the AOPDDR, described in Annex A of this document if these options are available. A clear statement shall be given when a zone(s) is described, whether its description is related to the detection zone(s) as defined in 3.4 or the combination of the detection zone(s) and the tolerance zone(s);		P
	eee) instructions to prevent an undetected passage of a person towards the hazard as described in Clause AA.5;		P
	fff) information about the behaviour of the AOPDDR in the presence of smoke and specular reflections;		P
	ggg) information on how the detection capability may be affected if the AOPDDR is used within an additional housing. For example, additional housings may have an influence on the detection capability and the detection zone;	No this situation	N/A
	hhh) if appropriate for the application(s), an indication on the floor of the detection zone should be recommended;	No this situation	N/A
	iii) instructions on how to document the setting of the detection zone(s) together with date, serial number of the AOPDDR and identification of the person responsible;	No this situation	N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	jjj) mounting restrictions according to 4.3.5 and 5.4.6.8.2, if the AOPDDR can be influenced during normal operation by an AOPDDR of identical design;		P
	kkk) information concerning external influences which may not be covered by this document and which may decrease the stated detection capability. Examples may include weld splatter, infra-red remote control devices, different fluorescent and stroboscopic light sources, snow, rain, pollution and thermal convection;		P
	lll) information concerning the need to check periodically the optical window(s) for damage (depending on the application);		P
	mmm) information concerning the need to check periodically the mounting of the AOPDDR for correctness and to check for possible misalignment of the detection zone(s) (depending on the application);		N/A
	nnn) information regarding the measures to be taken to avoid possible effects from optical radiation, if applicable;		P
	ooo) information as required by 4.1.6 if the AOPDDR possesses a zone(s) with limited detection capability;		N/A
	ppp) information regarding the avoidance of interference by incandescent light sources when required by 5.4.6.4.1 b). This information shall contain examples of light sources which may affect the AOPDDR in use and appropriate distances between the AOPDDR and these light sources;		
	qqq) information regarding the maximum speed in the worst-case direction within the detection zone of the AOPDDR of an object having the minimum detectable object size (see 4.2.12.3);		N/A
	rrr) information that an AOPDDR-2D having a minimum detectable object size greater than 117 mm shall not be used for direction of approach parallel to the detection zone according to ISO 13855:2010, 6.3;		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	sss) instruction that the minimum detectable object size shall be added to the safe distance calculations of ISO 13855 for AOPDDR-3D (see also Clause AA.4). This is because response time specifications assume that the object is entirely within the detection zone before it is detected;	AOPDDR-2D	N/A
	ttt) instructions that only qualified person should perform settings of the detection zone and/or other safety-related parameters.		P
Annex A	**Optional functions of the ESPE**		**-**
	Annex A of IEC 61496-1:2012 applies, except as follows:		-
	Addition:		-
A.9	**Setting the detection zone and/or other safety-related parameters**		**-**
A.9.1	Functional requirements		-
	The setting of the detection zone and/or other safety-related parameters shall not be possible without using a key, key-word or tool.	Not safety type ESPE	N/A
	NOTE For example, the tool can be a password protected software configuration program that is document of the AOPDDR.		-
	If the setting is carried out using a personal computer or equivalent fitted with untested dedicated hardware and/or software, a special procedure shall be used for setting the detection zone. This procedure shall be in accordance with appropriate standards (see also 4.2.11 of IEC 61496-1:2012). If the tool is software, only software authorized by the supplier shall be used.		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	The setting of safety-related parameters should only be performed by qualified persons. The procedure shall include measures to ensure that the input parameters are transmitted correctly and without corruption to the AOPDDR. This shall be applied for all safety-related settings, for example, the setting of the response time. The parameterization procedures shall conform to an appropriate standard (e.g. IEC 62061:2005, 6.11.2, or ISO 13849-1:2015, 4.6.4).		N/A
A.9.2	Verification		-
	The setting of a detection zone or other safety-relevant parameter(s) shall be verified as follows:		N/A
	a) verification of the correct setting function(s) for each configuration parameter (minimum, maximum and representative values); the possibility of differences between the detection zone as displayed on the screen of a configuration tool (for example, a personal computer) and the actual detection zone of the AOPDDR should be taken into account;		N/A
	b) verification that the configuration parameters are checked for plausibility, for example by use of invalid values, etc.;		N/A
	c) verification that the access to, and methods of, configuration by the user are in accordance with the requirements of corresponding standards;		N/A
	d) verification, in the case of detection zones that can be varied in size during operation, that the data/signals for determining the size of a detection zone are generated and processed in such a way that a single fault shall not lead to a loss of the safety function; verification that such a single fault is detected and causes the OSSDs to remain in the OFF-state or to go to the OFF-state within the response time of the AOPDDR.		N/A
A.10	Selection of multiple detection zones		N/A
A.10.1	Functional requirements		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	If an AOPDDR has more than one safety-related detection zone, a single fault shall not lead to an unintended change from one selected zone to another zone.		N/A
	For Type 3: In cases where a single fault which does not cause a failure to danger of the AOPDDR is not detected, the occurrence of a further fault internal to the AOPDDR shall not cause a failure to danger.		N/A
	Where the input signals are derived from device(s) external to the AOPDDR, this(those) device(s) should meet the relevant requirements of other appropriate standards (for example ISO 13849-1, IEC 61508 (all parts), IEC 62061).		N/A
	Single faults that prevent an intended change from one selected zone to another or prevent the activation of an additional safety-related detection zone shall cause the AOPDDR to go to a lock-out condition when a demand requires an activation of another zone or an activation of an additional zone. The specified response time(s) shall be maintained in this case.		N/A
	NOTE 1 It is possible that each zone has a different response time as specified by the supplier.		-
	If a detection zone is changed in size on-line for example by external inputs, the same requirement applies.		N/A
	The activation of the detection zones shall be monitored by the AOPDDR. The user shall have the possibility to configure the sequence of activation of the detection zones which is monitored by the AOPDDR. If an incorrect sequence of activation of the detection zones is detected, the AOPDDR shall respond by going to a lock-out condition.		N/A
	NOTE 2 The automatic selection of safety-related detection zones is not a muting function (as described in Clause A.7 of IEC 61496-1:2012).		-
A.10.2	Verification		-
	The functional requirements for the selection of multiple detection zones shall be verified as follows.		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	a) Verification that a single fault does not lead to an unintended change from one selected zone to another zone. Verification that a single fault does not prevent an intended change from one selected zone to another or prevent the activation of an additional safety-related detection zone.		N/A
	For Type 3: Verification that further faults will not lead to a failure to danger shall be carried out according to 5.3.4.		N/A
	b) Verification that common-mode failures cannot lead to a deactivation or variation of the detection zones.		N/A
	c) Verification that the specified response time of the AOPDDR is maintained in the case of switching between different detection zones.		N/A
	d) Verification that the user has the possibility to configure the sequence of activation of the detection zones which is monitored by the AOPDDR.		N/A
	e) Verification that the AOPDDR goes to the lock-out condition when the sequence of activation differs from that configured by the user.		N/A
	It is necessary to consider that persons can already be within the detection zone at the moment of switching between different detection zones.		N/A
A.11	Automatic setting of detection zones		N/A
A.11.1	Functional requirements		-
	If the AOPDDR has the possibility to automatically set the detection zone(s), the setting of the detection zone shall require verification by inspection of the detection zone(s) supported by manual checks.		N/A
	The automatic setting of a detection zone shall not be possible without using a tool. This tool can be, for example, a password protected software configuration program.		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	When determining the position accuracy of an automatically set detection zone, all conditions as listed in this document shall be taken into account, especially environmental interferences.		N/A
A.11.2	Verification		-
	The functional requirements for automatically setting a detection zone shall be verified by the following tests:		N/A
	a) tests according to A.9.2 a), b) and c);		N/A
	b) test whether the automatic setting of a detection zone(s) requires inspection of the detection zone(s) supported by manual checks;		N/A
	c) verification that a tool (for example, a password protected software configuration program) is necessary to enable automatic setting of a detection zone.		N/A
A.12	Reference boundary monitoring		N/A
A.12.1	Functional requirements		-
	Reference boundary monitoring is a measure that compares reference distances and distances measured by the AOPDDR. The reference distance is the distance between the AOPDDR and a boundary (for example a wall). The supplier has to specify the value of TZ_B (see Figure A.1).		N/A
	The AOPDDR shall monitor the reference distances and the value of TZ_B during operation. The OSSDs shall go to the OFF-state if the distance measurement values are outside the tolerances defined by the supplier.		N/A
	Figure A.1 shows the distribution of measurement values on an object to be detected. With $d > TZ + TZ_B$, the probability that an object is identified is sufficient. For simplification, Figure A.1 shows only the probabilistic aspects of the distribution of measurement values.		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	The sensing device of an AOPDDR shall be actuated and the OSSDs shall go to the OFFstate when a test piece in accordance with 4.2.13 is moving through the detection zone or the tolerance zone regarding the boundary (TZ_B) such that the direction of movement and the axis of the cylinder are orthogonal to the plane of the detection zone, at a speed of 1,6 m/s. Where the supplier states that an AOPDDR can be used to detect objects moving at speeds greater than 1,6 m/s, this requirement shall be met at the stated maximum speed. When the OSSD(s) go to the OFF-state, they shall remain in the OFF-state while the test piece is present in the detection zone.		N/A
	NOTE The purpose of this requirement is to ensure that the OSSDs go to the OFF-state when a person or part of a person passes through the detection zone.		-
	Figure A.2 shows an example for using an AOPDDR as a whole body trip device. Figure A.3 shows an example for using an AOPDDR as parts of a body trip device. The value of a shown in Figure A.3 shall be small enough to ensure detection of the test piece.		N/A
A.12.2	Verification		-
	Verify that:		-
	a) the OSSDs go to the OFF-state if the distance measurement values are outside the range defined by the supplier for the reference boundary monitoring;		N/A
	b) the stated detection capability is not exceeded;		N/A
	c) the accompanying documents contain the information necessary to ensure that persons cannot pass undetected through the detection zone;		N/A

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EN IEC 61496-3			
Clause	Requirement + Test	Result - Remark	Verdict
	d) the sensing device is actuated and the OSSDs go to and remain in the OFF-state when a test piece defined in 4.2.13.3 is moving through the detection zone or the tolerance zone regarding the boundary (TZ_B) such that the direction of movement and the axis of the cylinder are orthogonal to the plane of the detection zone at a speed of 1,6 m/s. Where the supplier states that an AOPDDR can be used to detect objects moving at speeds greater than 1,6 m/s, verification shall be carried out at the stated maximum speed.		N/A
Annex AA	Examples of the use of an AOPDDR in different applications	informative	-
Annex BB	Relationship between position accuracy and probability of detection	informative	-

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Test equipment list

No.	Instrument	Model	Calibration date to
1	Flashing Light Source	PN-01F	2023/04/24
2	Fluorescent Light Source	PHILIPS	2023/07/14
3	Incandescent Light Source	SYLVANIA	2023/07/14
4	Illumination Photometer	T10A	2022/11/19
5	Electric Vibration Test System	M232A	2023/01/08
6	Hydrostatic Horizontal Slider	GT600M	2023/01/08
7	Comprehensive water supply test box	ZSH	2023/07/11
8	Centralized Control Box	JZK	2023/07/11
9	Hand-held Sprinkler	SSS	2023/07/11
10	Rotary Table	XZT	2023/07/11
11	Sand Dust Test Chamber	XH351	2023/07/11
12	Electrostatic Discharge Tester	NSG 437	2022/12/14
13	Conduction Tolerance Tester	NSG4070	2023/04/07
14	2P/2W coupling decoupling network	CDN MO16S	2023/04/07
15	Surge Burst Tester	NSG 3040	2023/04/07
16	Coupled Decoupling Network	CDN 3063	2023/04/07
17	Capacitive Coupling Clip	CDN 3425	2023/04/07
18	SAC	SAC 3m	2023/11/17
19	Signal Generator	SMB 100A 106	2023/04/07
20	Field Intensity Meter	EP601	2022/12/20
21	Log Periodic Antenna	STLP 9128 D	2022/11/20
22	AVO Meter	FLUKE8846A	2023/07/15
23	Electronic Recording Instrument	174H	2023/07/10
24	Oscilloscope	DS4034	2023/03/30
25	Steel Tap	7.5m	2023/01/11
26	Climate Test Chamber	VC^3 4060	2023/07/11
27	Climate Test Chamber	VC^3 7150	2023/07/11
28	Pendulum Test Device	BCT-1	2023/07/08

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29	Spring Impact Hammer	PTL	2022/12/12
30	Voltage Drop Simulator Battery Supply Simulator And DC Voltage Source APS 60C30 Series	APS60C30	2023/01/05
31	High Precision Program DC Power Supply	DCP60C30	2023/01/05
32	Laser Range Finder	PD-1	2023/04/01
33	Second Chronograph	PC396	2023/07/15
34	Color Luminance Meter	CS200	2022/12/09
35	ESD Simulator	DIOT 509030	2023/01/25
36	Digital Pressure Meter	DYM3	2023/01/31
37	Temperature & Humidity Recorder	TH603	2022/10/13

--- End of Test Report ---

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